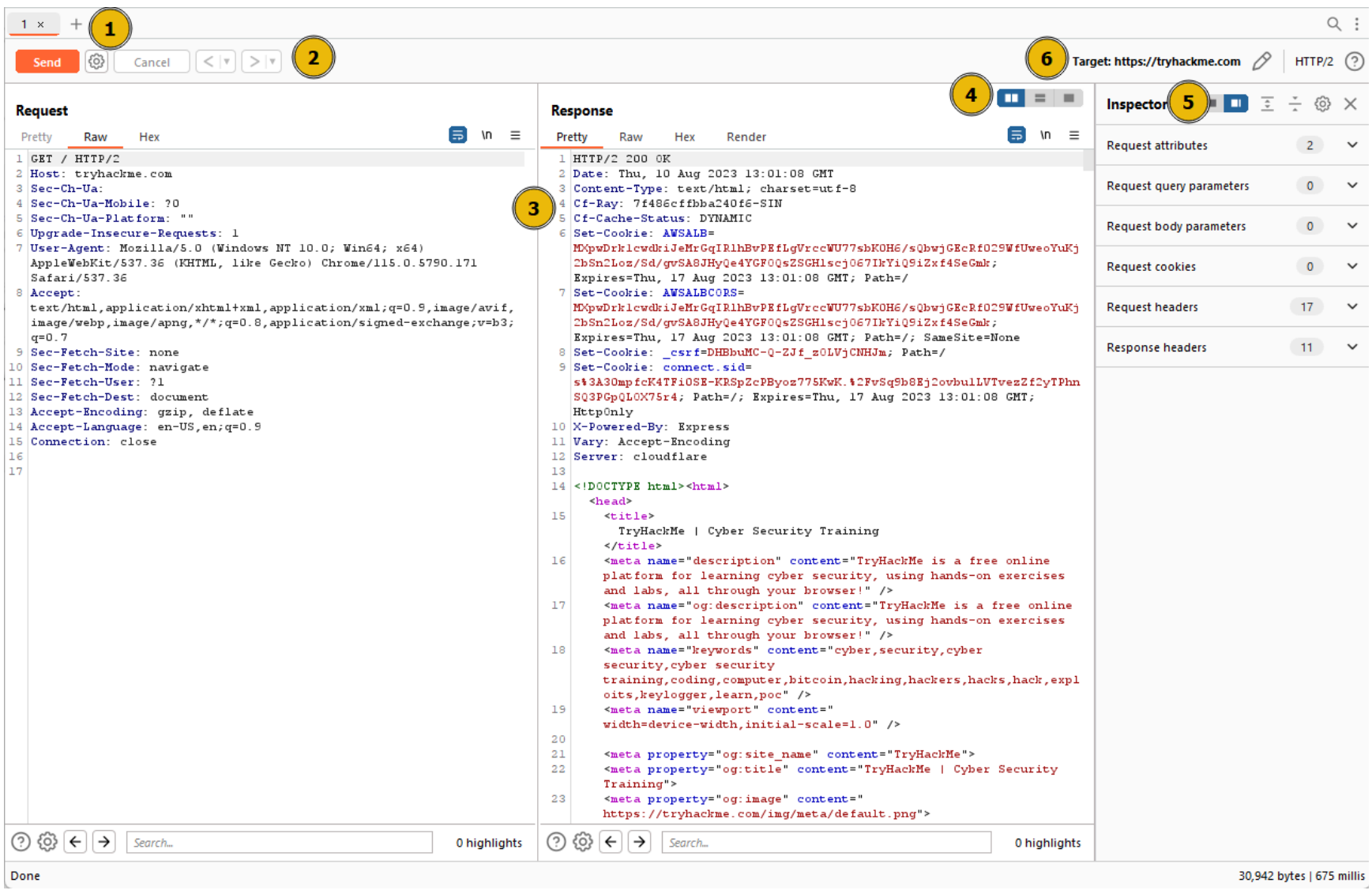
BURPSUITE REPEATER

Before using Burp Suite Repeater, let's familiarize ourselves with its purpose and functionality.

In essence, Burp Suite Repeater enables us to modify and resend intercepted requests to a target of our choosing. It allows us to take requests captured in the Burp Proxy and manipulate them, sending them repeatedly as needed. Alternatively, we can manually create requests from scratch, similar to using a command-line tool like cURL.

The ability to edit and resend requests multiple times makes Repeater invaluable for manual exploration and testing of endpoints. It provides a user-friendly graphical interface for crafting request payloads and offers various views of the response, including a rendering engine for a graphical representation.

The Repeater interface consists of six main sections, as depicted in the annotated diagram below:



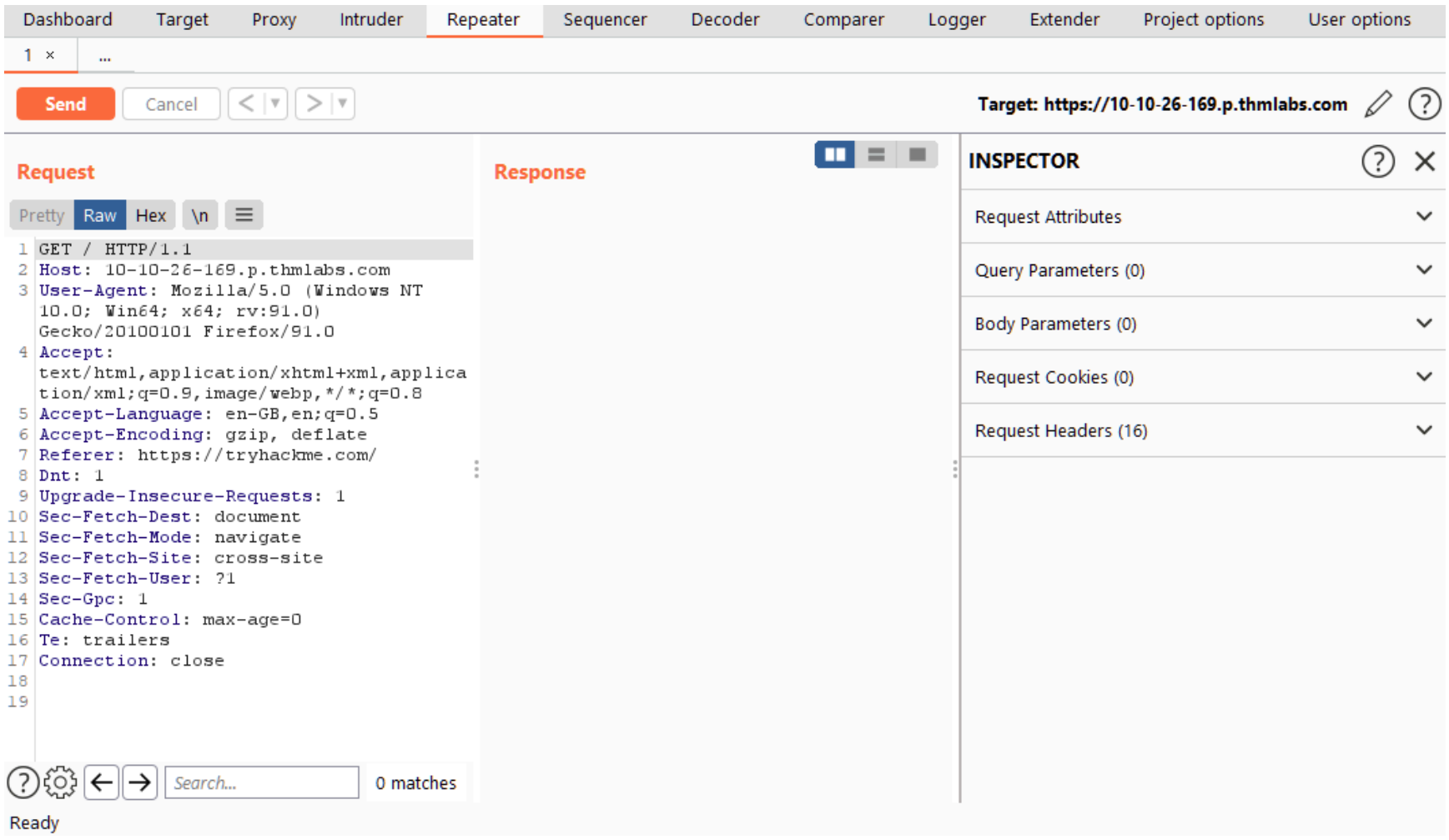
1. **Request List**: Located at the top left of the tab, it displays the list of Repeater requests. Multiple requests can be managed simultaneously, and each new request sent to Repeater will appear here.
2. **Request Controls**: Positioned directly beneath the request list, these controls allow us to send a request, cancel a hanging request, and navigate through the request history.
3. **Request and Response View**: Occupying the majority of the interface, this section displays the **Request** and **Response** views. We can edit the request in the Request view and then forward it, while the corresponding response will be shown in the Response view.
4. **Layout Options**: Located at the top-right of the Request/Response view, these options enable us to customize the layout of the Request and Response views. The default setting is a side-by-side (horizontal) layout, but we can also choose a vertical layout or combine them in separate tabs.
5. **Inspector**: Positioned on the right-hand side, the Inspector allows us to analyze and modify requests in a more intuitive manner than using the raw editor. We will explore this feature in a later task.
6. **Target**: Situated above the Inspector, the Target field specifies the IP address or domain to which the requests are sent. When requests are sent to Repeater from other Burp Suite components, this field is automatically populated.

We know what the application's interface looks like at this point, but how can we effectively utilize it?

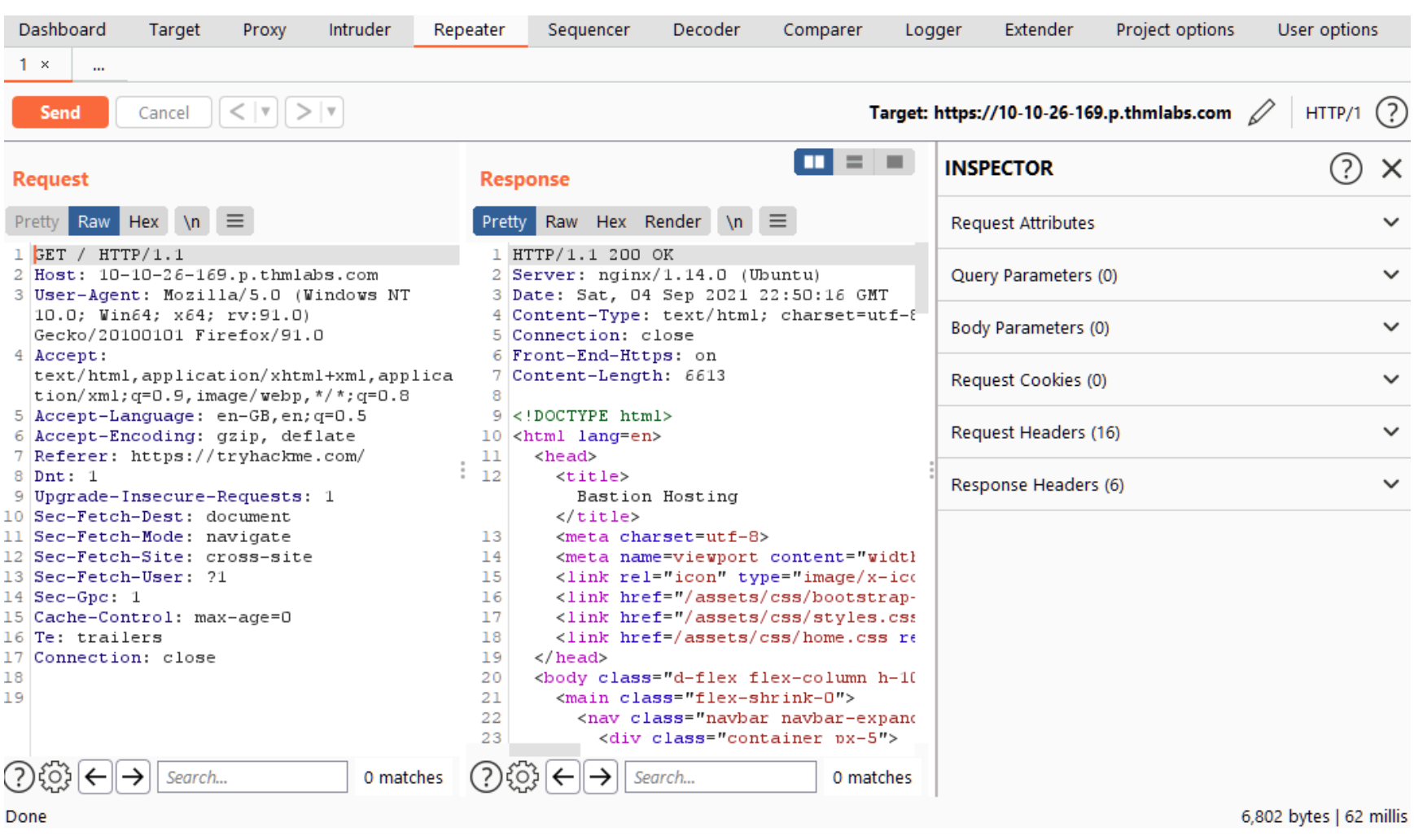
While manual request crafting is an option, it is more common to capture a request using the Proxy module and subsequently transmit it to Repeater for further editing and resending.

Once a request has been captured in the Proxy module, we can send it to Repeater by either right-clicking on the request and selecting **Send to Repeater**, or by utilizing the keyboard shortcut `Ctrl + R`.

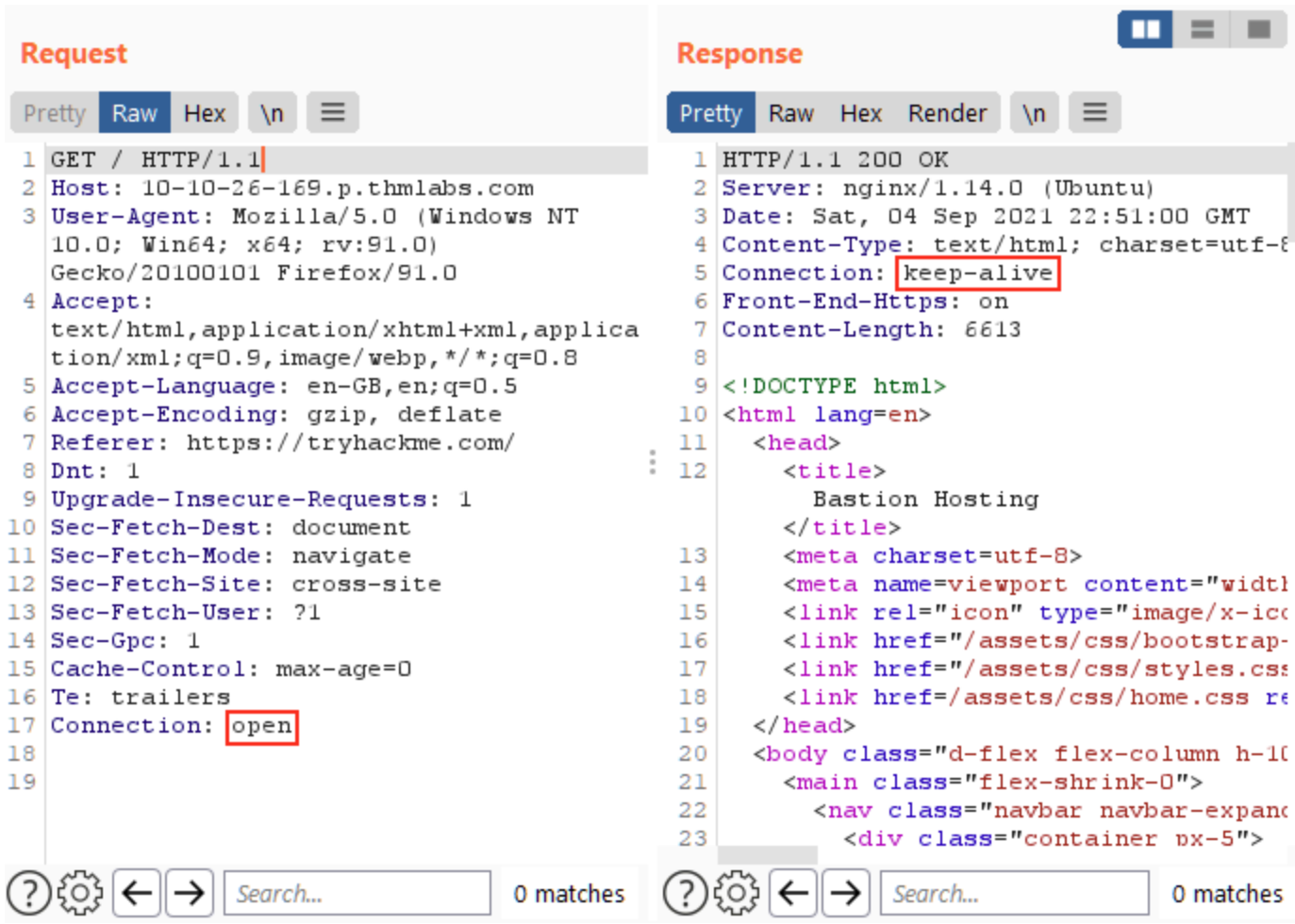
Shifting our focus back to Repeater, we can observe that our captured request is now accessible in the Request view:



Both the Target and Inspector sections now display relevant information, albeit we are currently lacking a response. Upon clicking the **Send** button, the Response view swiftly populates:



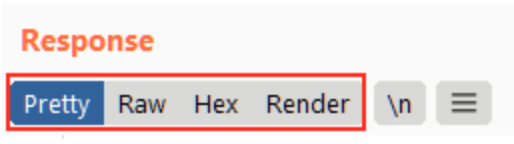
Should we wish to modify any aspect of the request, we can simply type within the Request view and press **Send** once again. This action will update the Response view on the right accordingly. For instance, altering the **Connection** header to "open" instead of "close" yields a response with a **Connection** header containing the value "keep-alive":



Furthermore, we can utilize the history buttons situated to the right of the Send button to navigate through our modification history, allowing us to move forward or backwards as needed.

Repeater provides us with various request and response presentation options, ranging from hexadecimal output to a fully rendered page.

To explore these options, we can refer to the section located above the response box, where the following four view buttons are available:



We are presented with the following display choices:

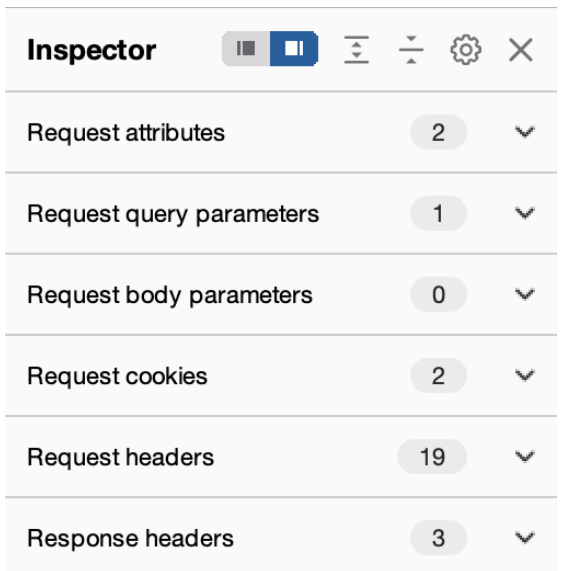
1. **Pretty**: This is the default option, which takes the raw response and applies slight formatting enhancements to improve readability.
2. **Raw**: This option displays the unmodified response directly received from the server without any additional formatting.
3. **Hex**: By selecting this view, we can examine the response in a byte-level representation, which is particularly useful when dealing with binary files.
4. **Render**: The render option allows us to visualize the page as it would appear in a web browser. While not commonly utilised in Repeater, as our focus is usually on the source code, it still offers a valuable feature. For most scenarios, the **Pretty** option is generally sufficient. However, it is beneficial to be acquainted with the usage of the other three options.

Adjacent to the view buttons, on the right-hand side, we find the **Show non-printable** characters button (`\n`). This functionality enables the display of characters that may not be visible with the **Pretty** or **Raw** options. For example, each line in the response typically ends with the characters `\r\n`, representing a carriage return followed by a new line. These characters play an important role in the interpretation of HTTP headers.

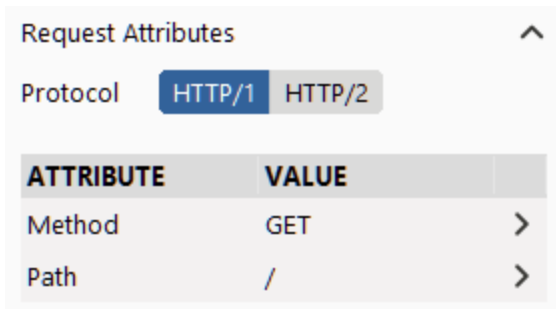
While not mandatory for most tasks, this option can prove advantageous in certain situations.

Inspector is a supplementary feature to the Request and Response views in the Repeater module. It is also used to obtain a visually organized breakdown of requests and responses, as well as for experimenting to see how changes made using the higher-level Inspector affect the equivalent raw versions.

Inspector can be utilized both in the Proxy and Repeater module. In both instances, it is situated on the far-right side of the window, presenting a list of components within the request and response:



Among these components, the sections pertaining to the request can typically be modified, enabling the addition, editing, and removal of items. For instance, in the **Request Attributes** section, we can alter elements related to the location, method, and protocol of the request. This includes modifying the desired resource to retrieve, changing the HTTP method from GET to another variant, or switching the protocol from HTTP/1 to HTTP/2:



Other sections available for viewing and/or editing include:

1. **Request Query Parameters:** These refer to data sent to the server via the URL. For example, in a GET request like `https://admin.tryhackme.com/?redirect=false`, the query parameter **redirect** has a value of "false".
2. **Request Body Parameters:** Similar to query parameters, but specific to POST requests. Any data sent as part of a POST request will be displayed in this section, allowing us to modify the parameters before resending.
3. **Request Cookies:** This section contains a modifiable list of cookies sent with each request.
4. **Request Headers:** It enables us to view, access, and modify (including adding or removing) any headers sent with our requests. Editing these headers can be valuable when examining how a web server responds to unexpected headers.
5. **Response Headers:** This section displays the headers returned by the server in response to our request. It cannot be modified, as we have no control over the headers returned by the server. Note that this section becomes visible only after sending a request and receiving a response.

While the textual representation of these components can be found within the Request and Response views, Inspector's tabular format provides a convenient way to visualise and interact with them. Experimenting with header additions, removals, and edits in Inspector helps grasp how the corresponding raw version changes in response.